Exams without clashes

Optimization · Level 3 · Lesson 2



Today we will

- Draw exam clashes as a map and colour it with few slots.
- Prove the fewest slots with a clique, or a solver’s OPTIMAL.
- Weigh a fairness rule against a short exam week.

Remember

- Exams are dots, shared pupils are lines, and slots are colours.
- Quick rules find a plan; a clique or the solver proves the fewest.
- The fewest slots isn’t always the kindest: decide the trade-off together.

The chalk clash map

1. Warm-up (5 min): Who has ever had two exams at once?
2. Teach (8 min): Dots, lines and colours.
3. Draw and colour (15 min): Groups colour the clash map.
4. Prove it (7 min): Find a group that all clash.
5. Wrap-up (5 min): Add “no two in a row” and count again.

Use the lab’s eight exams if you have no real list: maths, English, science, Arabic, history, geography, art and French, with the six pupils in the “Exam slots without clashes” lab.

Let’s discuss

1. How do we know nobody can use fewer slots?
2. Who had the hardest exam days in our plan?
3. Is a shorter exam week worth three tired pupils?

Exit ticket

- Draw a clash map of four exams that needs exactly three slots, and say why.